

## HELIOS OBSERVABILITY PLATFORM

**Service:** payment-service**Severity:** low

# Incident Report: anomaly\_payment-service\_1779388033

---

**Service:** payment-service**Detected:** 2026-05-21T18:27:13.137486**Severity:** LOW**Model confidence:** 0.95

## Executive summary

The payment-service is experiencing a significant latency spike and elevated error rates. The issue is characterized by recurring database timeouts, impacting request performance at the 95th and 99th percentiles.

## Root cause hypothesis

The payment-service is suffering from a database-level bottleneck, evidenced by multiple 'DB timeout' log entries. The anomaly is primarily driven by p99\_latency\_ms and p95\_latency\_ms, which indicate that a subset of requests is experiencing severe delays, while the high latency\_std suggests significant variance in response times consistent with contention or locking issues in the database layer.

## Top contributing features (SHAP)

| Feature        | Value    | SHAP   | Direction      |
|----------------|----------|--------|----------------|
| p99_latency_ms | 5191.900 | -2.152 | toward_anomaly |
| latency_std    | 1178.902 | -1.804 | toward_anomaly |
| p95_latency_ms | 3682.628 | -1.693 | toward_anomaly |

## Recommended actions

### Immediate

- **Check database connection pool health and active transaction counts.** — The presence of repeated 'DB timeout' log events suggests the service is unable to acquire database connections or queries are hanging.

### Short Term

- **Review recent database schema changes or query performance regressions.** — The high p99\_latency\_ms and latency\_std values indicate that specific requests are stalling, likely due to inefficient query execution.

### Long Term

- **Implement circuit breakers for database-dependent calls.** — Preventing cascading failures during database timeouts will improve service resilience against similar latency spikes in the future.

## Monitoring follow-up

- payment\_service\_db\_timeout\_count
- payment\_service\_p99\_latency
- database\_connection\_pool\_utilization
- database\_query\_execution\_time

*Generated by Helios at 2026-05-21T18:27:40+00:00*

---

This is an automated incident report generated by Helios Observability Platform  
For questions or concerns, contact your DevOps team